

✿ XML + DTD + XSD – Complete Exam Cheat Sheet

1 XML Basics

Color suggestion: Dark Blue for headings, Black for text, Gray for notes

- **What is XML?**
 - Stores and transports data (semi-structured)
 - Hierarchical (tree) structure
 - Used to represent data flexibly compared to HTML (display) or DB (tables)
- **Rules for well-formed XML:**
 1. Only **one root element**
 2. Tags must **close**
 3. Proper **nesting**
 4. **Case-sensitive**

Example:

```
<?xml version="1.0" encoding="UTF-8"?>
<root>
  <name>Sarah</name>
  <age>20</age>
</root>
```

- **Elements vs Attributes**

```
<student id="S1" level="L3">
  <name>Sarah</name>
  <age>20</age>
</student>
```

- Elements → main data
 - Attributes → metadata
- **Empty element:** <tag/>
- **Comments:** <!-- This is a comment -->
- **Special chars:** < → <; > → >; & → &
- **CDATA (ignore parsing):** <![CDATA[<hello> not a tag]]>

2 DTD – Write Code

Color suggestion: Dark Red headings, Black text, Blue code

- **Role:** Define structure of XML (elements, attributes, order, constraints)

Basic Syntax:

```
<!ELEMENT name content>  
<!ATTLIST element attribute TYPE DEFAULT>
```

- **Content types:**
 - #PCDATA → text only
 - (A, B) → ordered children
 - (A | B) → choice
 - EMPTY → no content
- **Operators:**

Symbol Meaning

,	order
	choice
?	0 or 1
*	0+
+	1+

Example DTD:

```
<!ELEMENT students (student+)>  
<!ELEMENT student (name, age, phone*)>  
<!ELEMENT name (#PCDATA)>  
<!ELEMENT age (#PCDATA)>  
<!ELEMENT phone (#PCDATA)>  
<!ATTLIST student id ID #REQUIRED  
                  level (L1 | L2 | L3) #IMPLIED>
```

- **DTD Types:**
 - Internal → inside XML file
 - External → reference to .dtd file
 - Mixed → combination

3 XSD – Theory Questions

Color suggestion: Dark Green headings, Black text

- **Namespace:** avoids element name conflicts

```
xmlns:xs="http://www.w3.org/2001/XMLSchema"
```

- **Use of XSD:** define and validate XML structure
 - Elements
 - Attributes
 - Order (sequence)
 - Occurrences (min/max)
 - Data types

- **Advantages:**
 - Written in XML → easier
 - Supports types (int, string...)
 - Strong validation
 - Supports namespaces
 - Extensible
- **Differences vs DTD:**

Feature	DTD	XSD
Syntax	Not XML	XML
Data Types	✗ none	☑ int, string, etc.
Validation	Limited	Strong
Namespaces	✗	☑
Extensibility	✗	☑

- **Important Keywords:**
 - `xs:schema` → root
 - `xs:element` → define element
 - `xs:complexType` → children + attributes
 - `xs:simpleType` → simple value
 - `xs:sequence` → order
 - `xs:attribute` → attribute
 - `minOccurs` / `maxOccurs` → occurrences
 - `xs:string`, `xs:int` → types

4 Full Example (XML + DTD + XSD)

Color suggestion: Purple headings, Black text, Blue code

XML + Internal DTD

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE students [
<!ELEMENT students (student+)>
<!ELEMENT student (name, age, phone*)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT age (#PCDATA)>
<!ELEMENT phone (#PCDATA)>
<!ATTLIST student id ID #REQUIRED>
]>
<students>
  <student id="S1" level="L3">
    <name>Sarah</name>
    <age>20</age>
    <phone>0550000000</phone>
    <phone>0660000000</phone>
  </student>
  <student id="S2">
    <name>Ali</name>
```

```
    <age>22</age>
  </student>
</students>
```

XSD Equivalent

```
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="students">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="student" maxOccurs="unbounded">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="name" type="xs:string"/>
              <xs:element name="age" type="xs:int"/>
              <xs:element name="phone" type="xs:string" minOccurs="0"
maxOccurs="unbounded"/>
            </xs:sequence>
            <xs:attribute name="id" type="xs:string" use="required"/>
            <xs:attribute name="level" use="optional">
              <xs:simpleType>
                <xs:restriction base="xs:string">
                  <xs:enumeration value="L1"/>
                  <xs:enumeration value="L2"/>
                  <xs:enumeration value="L3"/>
                </xs:restriction>
              </xs:simpleType>
            </xs:attribute>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```